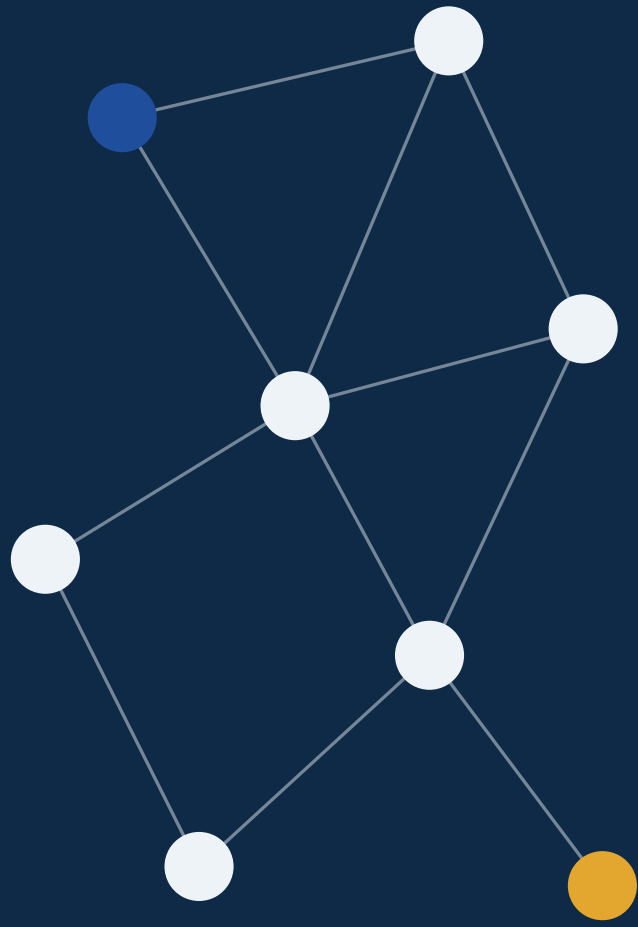
Performance Analysis of AODV in a MANET Using NS-3

with a Comparison against OLSR and DSDV



Alternate Assessment · Network Programming (23CS7PCNWP)

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Why routing in a MANET is hard

A mobile ad hoc network has no access point or router: every node forwards packets for the others, and the routing protocol must keep finding paths as nodes move.



Partitions

A sparse network splits into parts. If no path exists, no protocol can deliver.



Mobility

Moving nodes break links. Routes must be detected as broken and repaired quickly.



Shared channel

Data and routing control packets compete for one radio channel, hop after hop.

Objectives

1. Implement a MANET in ns-3.48 with a UDP client–server application
2. Measure AODV as node density, speed and traffic load change
3. Compare AODV with OLSR and DSDV on identical topologies
4. Measure routing overhead, jitter and packets dropped at the source
5. Validate the results and report 95% confidence and significance

420

simulations

14 settings × 3 protocols × 10 random topologies

Three protocols, two approaches

AODV

Reactive · distance vector

Port

UDP port 654

Routes

Finds a route only when needed; buffers packets meanwhile

Control messages

RREQ, RREP, RERR; Hello every 1 s

OLSR

Proactive · link state

Port

UDP port 698

Routes

Routes always in the table; drops packets without a route

Control messages

HELLO every 2 s; TC every 5 s via multipoint relays

DSDV

Proactive · distance vector

Port

UDP port 269

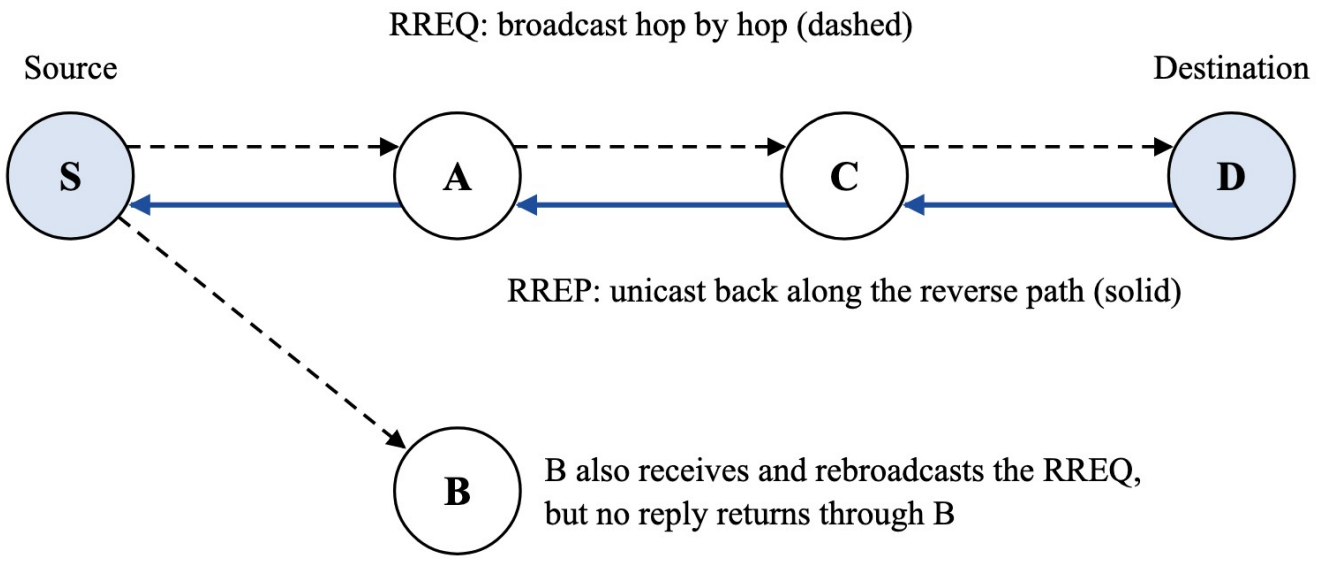
Routes

Routes always in the table; buffers until a route appears

Control messages

Full update every 15 s plus triggered updates

AODV finds routes on demand



RREQ The source broadcasts a route request; every node records a reverse route and rebroadcasts it.

RREP The destination replies by unicast along the reverse path, installing the forward route.

RERR When a link on an active route breaks, a route error tells the nodes using it.

Seq. no. Destination sequence numbers keep routes fresh and loop-free.

Research gap and problem statement

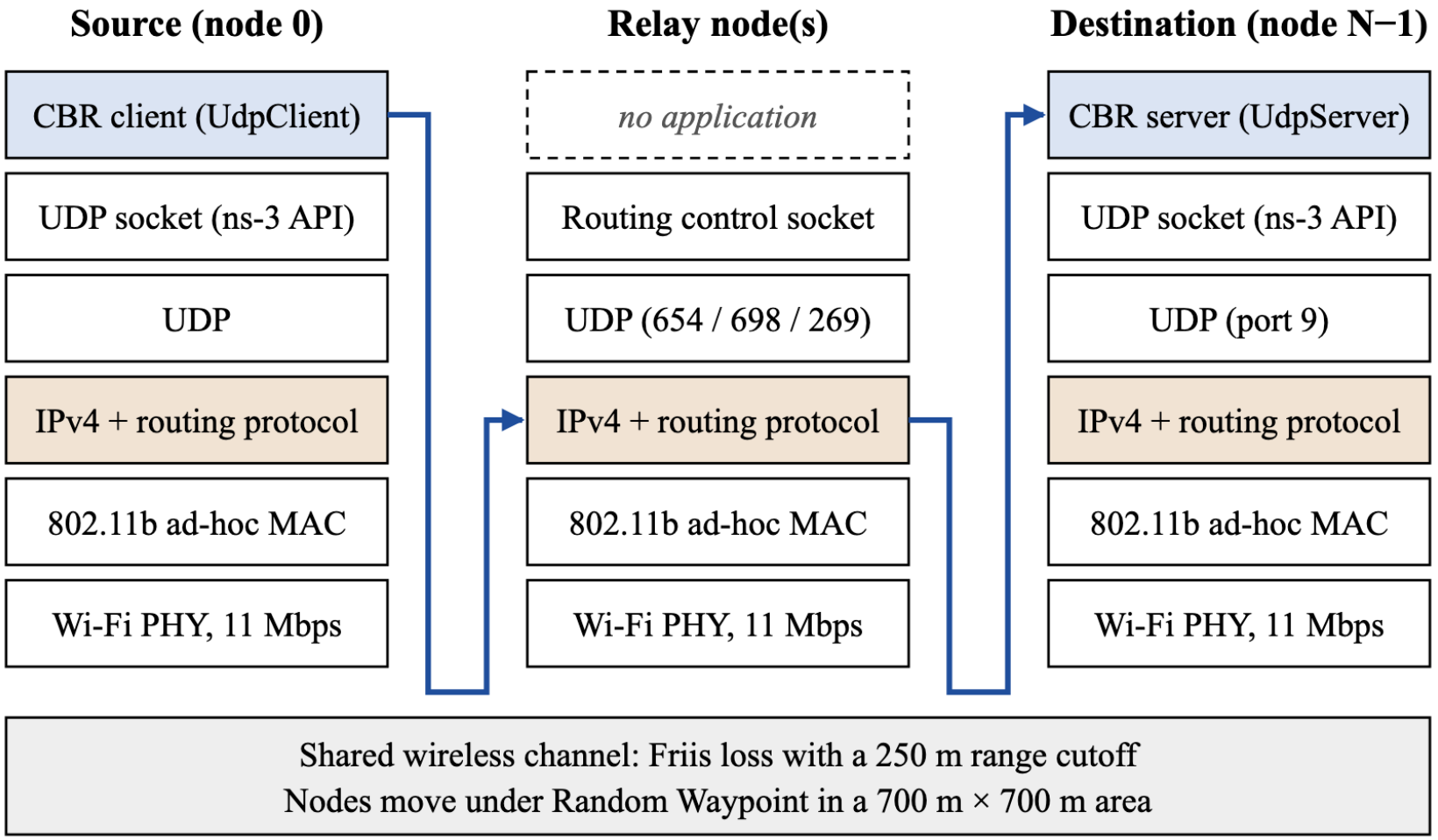
Research gap

- Results change with density, mobility and load, so rankings depend on the scenario
- Many MANET simulations cannot be repeated or report no spread
- New AODV variants are all judged against standard AODV, so a reproducible baseline comparison is needed

Problem statement

Implement a MANET in ns-3.48 with a UDP client–server application and determine, with repeated runs and statistical confidence, how AODV's delivery, delay, jitter and overhead change with density, speed and load, and how it compares with OLSR and DSDV on the same topologies.

System architecture



- Every node runs the same stack: 802.11b ad-hoc Wi-Fi, IPv4 with the routing protocol, UDP and the ns-3 socket API
- Relays forward data at the IPv4 layer using the protocol's routes
- FlowMonitor measures the data flow; packet traces count control packets

Socket programming in the simulation

BSD socket call	ns-3 Socket API	Use in this project
`socket()`	`Socket::CreateSocket(node, UdpSocketFactory)`	Client, server and routing sockets
`bind()`	`Bind(InetSocketAddress(addr, port))`	Server port 9; routing 654 / 698 / 269
`connect()`	`Connect(InetSocketAddress(dst, 9))`	Client fixes the server address
`send()` / `sendto()`	`Send(packet)` / `SendTo(packet, 0, dst)`	Data every 1/λ s; routing messages
`recvfrom()`	`RecvFrom(from)` in a receive callback	Server counts packets; routing reads updates
`setsockopt(SO_BROADCAST)`	`SetAllowBroadcast(true)`	RREQ, HELLO and update broadcasts

ns-3 has no kernel, so these are C++ calls on simulated sockets. Nothing may block in a discrete-event simulator, so reception is event-driven through SetRecvCallback() instead of a blocking recvfrom().

Experimental setup

Simulator	ns-3.48 (C++)
Area, nodes	700 m × 700 m, 10 nodes
Wi-Fi	802.11b ad-hoc, 11 Mbps, 250 m range
Mobility	Random Waypoint, 5 m/s, 0 s pause
Traffic	UDP CBR, 512 B, 4 packets/s, node 0 → N−1
Time	60 s (traffic 2–59 s)
Repetitions	10 runs; run k = same topology for all protocols

A **Number of nodes**

5, 10, 15, 20

B **Node speed**

1, 5, 10, 15 m/s

C **Traffic load**

10 – 800 packets/s

One parameter changes at a time; the others keep their defaults.

What we measured

Packet delivery ratio

Received ÷ sent data packets

Throughput

Delivered bits per second at the IP layer

End-to-end delay

Mean transit time of delivered packets

Jitter

Mean change in delay between consecutive packets

Routing overhead

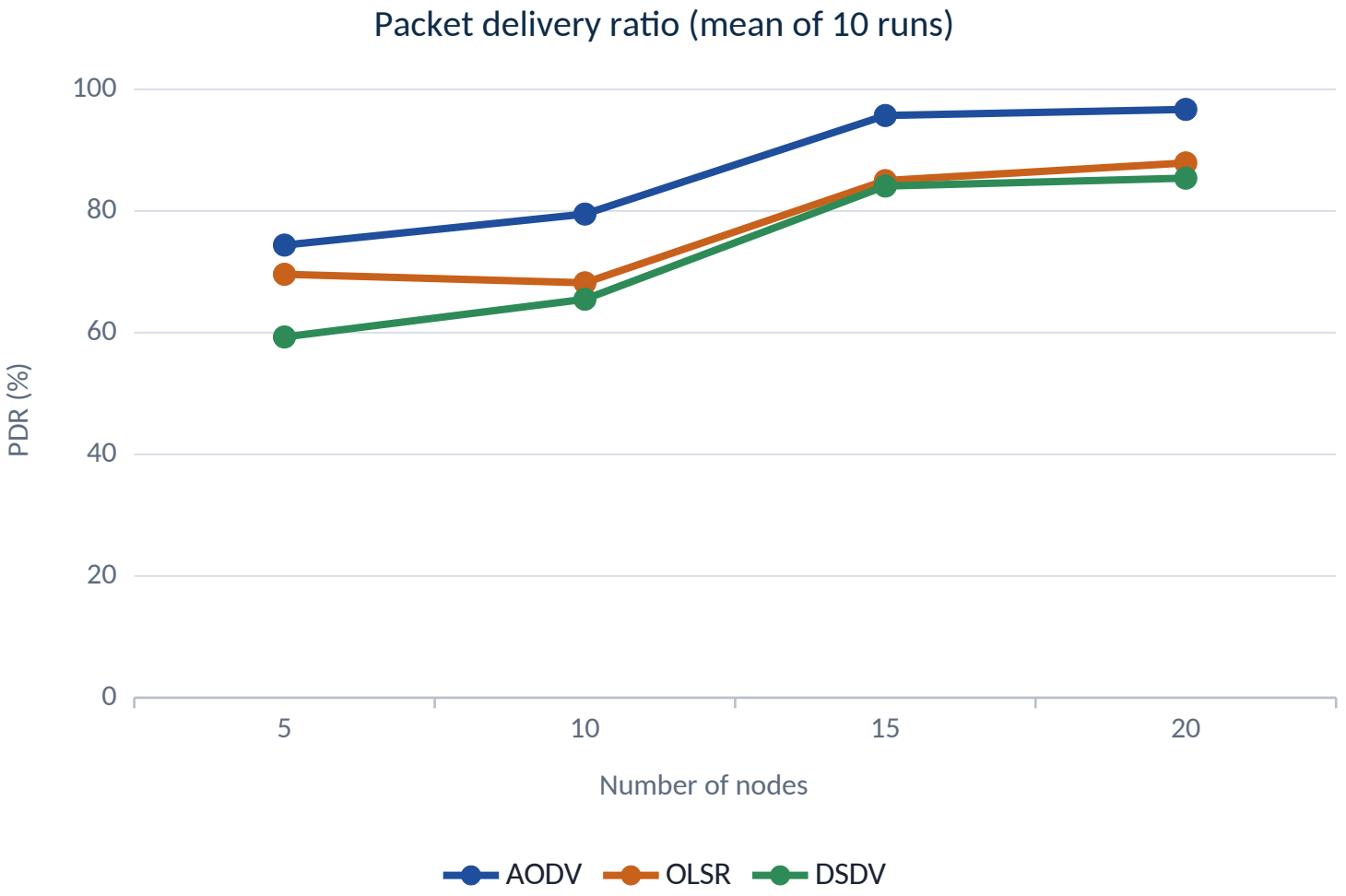
Control packets sent by all nodes; per delivered packet = NRL

Source drops

Packets dropped at the source for lack of a route

Overhead is counted with the Ipv4L3Protocol Tx trace because FlowMonitor ignores broadcast packets; sent data packets are counted at the client because OLSR drops packets before FlowMonitor sees them.

More nodes help every protocol; AODV stays ahead



AODV

74% → 97%

OLSR

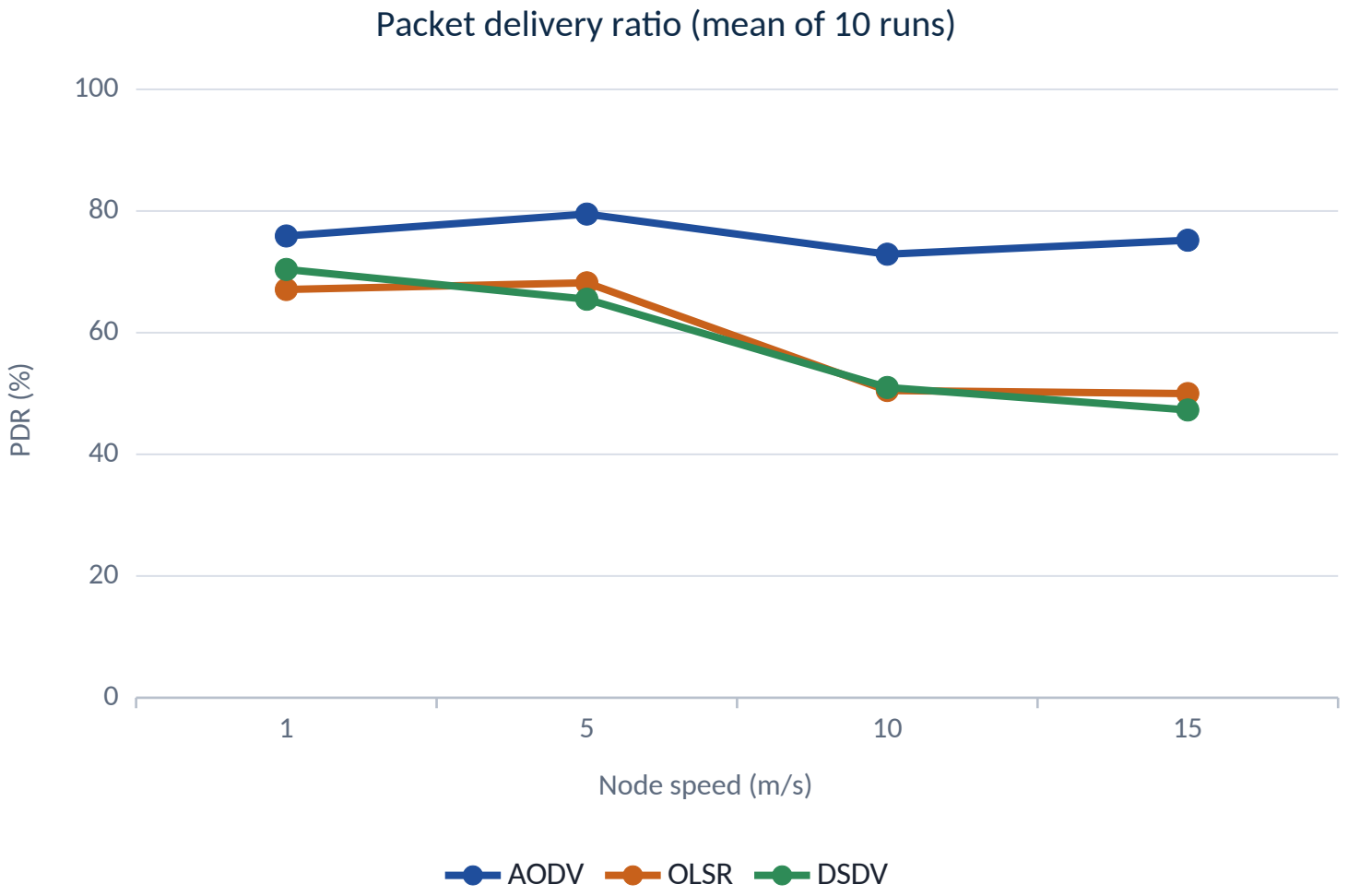
70% → 88%

DSDV

59% → 85%

Denser networks rarely split, so a path almost always exists.

Under mobility, AODV holds its delivery



At 15 m/s

75%

AODV delivery

50%

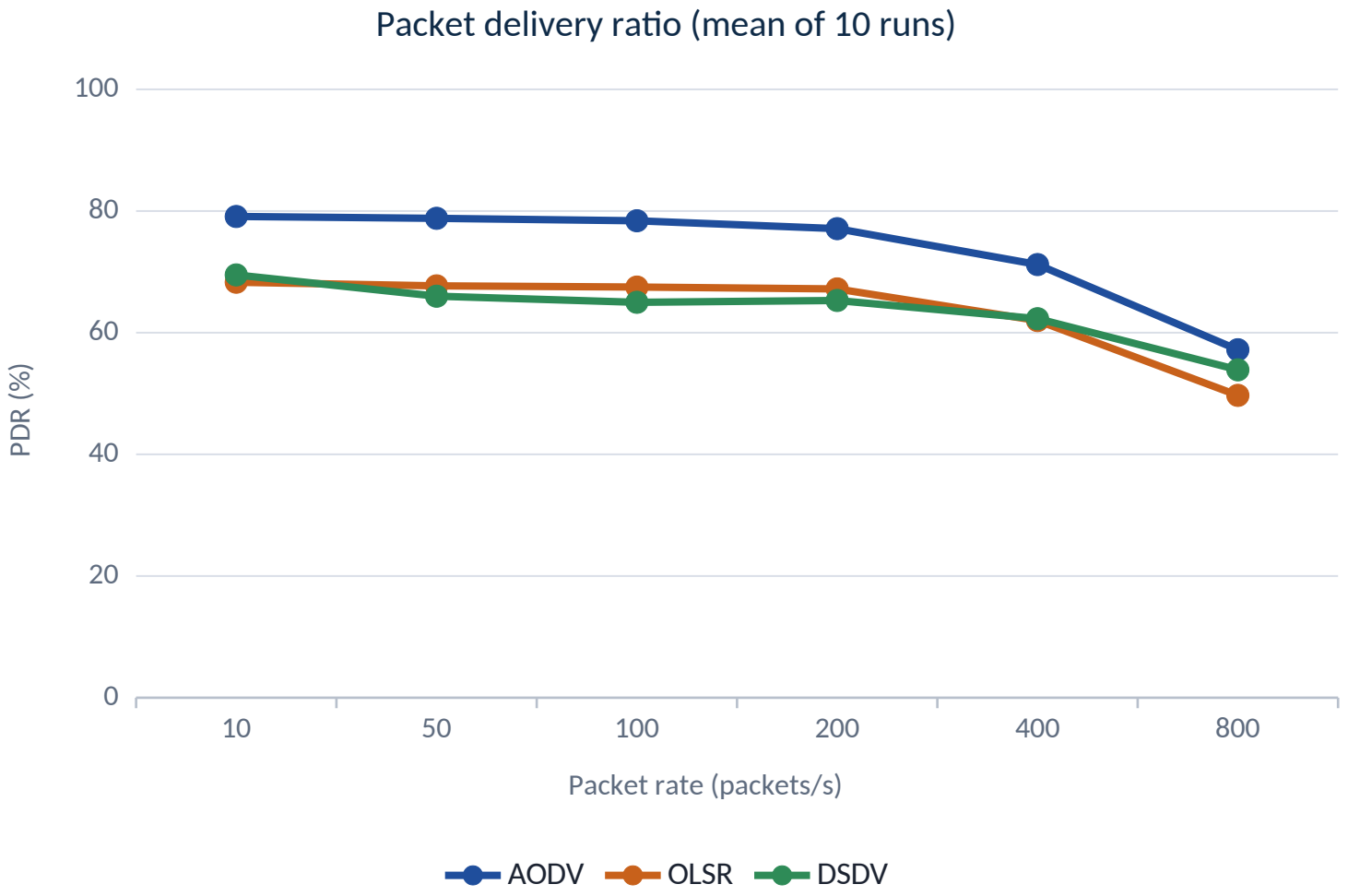
OLSR delivery

47%

DSDV delivery

Proactive tables refresh every 2–15 s, so fast nodes leave them stale; AODV repairs routes on demand.

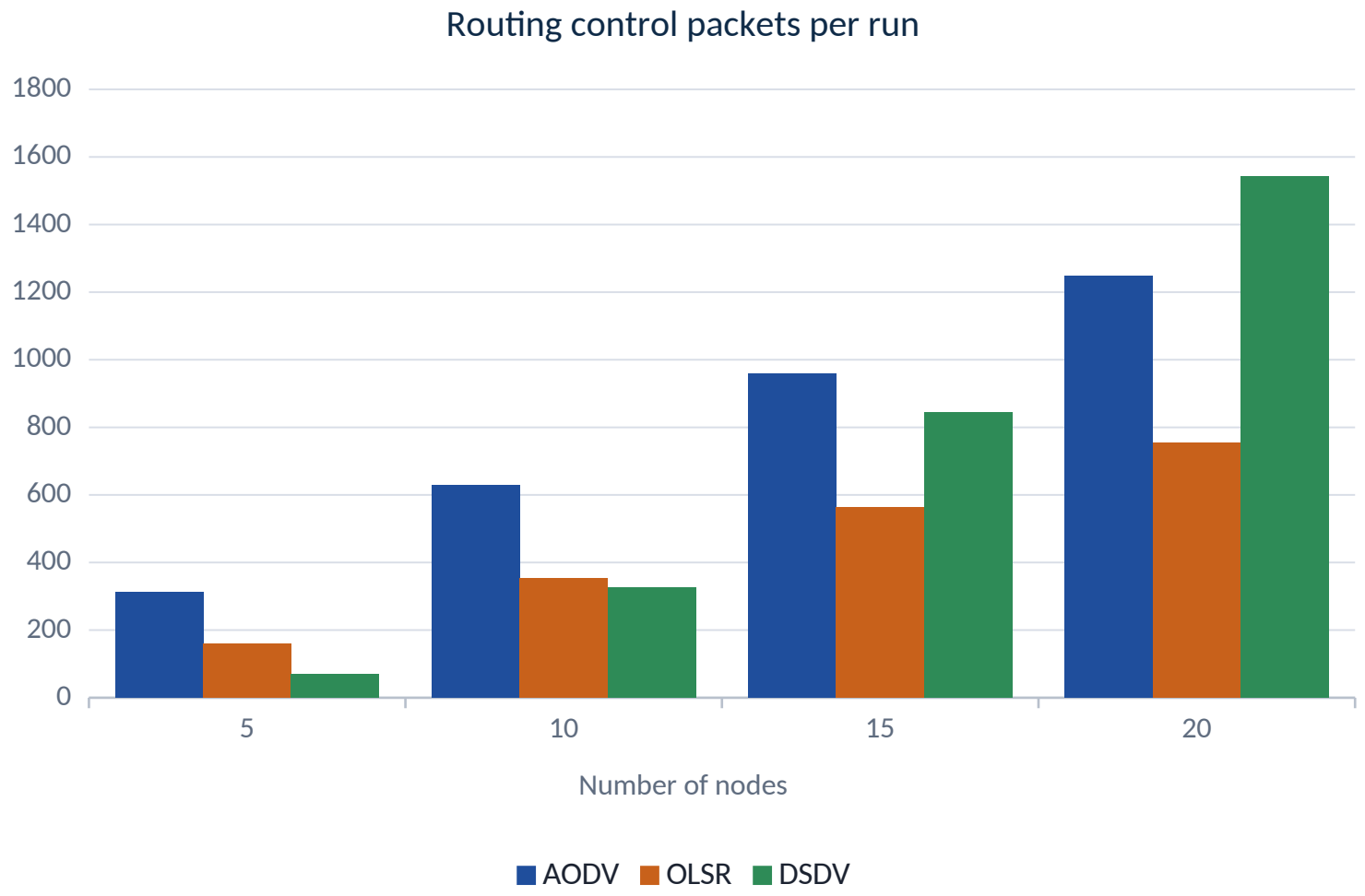
Heavy load congests every protocol



Up to 200 packets/s
Throughput follows the offered load; delivery stays at its low-load level

At 800 packets/s
PDR falls to 57% (AODV), 54% (DSDV) and 50% (OLSR); delay rises to 170–240 ms

The price: routing overhead and delay



≈ 2×

AODV control packets vs OLSR and DSDV (10 nodes: 630 vs 353 / 325)

41 ms

AODV mean delay at 10 nodes, vs 2 ms for OLSR: AODV buffers packets during discovery

Comparative summary

77.9%

AODV: mean PDR over 13 settings

66.2%

OLSR: mean PDR over 13 settings

65.0%

DSDV: mean PDR over 13 settings

AODV delivers significantly more in 12 of 13 settings than OLSR and 9 of 13 than DSDV, and never significantly less

AODV sends about twice the routing overhead and has a higher mean delay and jitter

Paired by topology, so differences are significant despite large run-to-run spread

The results are validated

Regression

All 140 AODV runs identical to the original study

Packets sent

57 s × rate in all 420 runs

Loss and PDR

Loss = 100 − PDR in all 420 runs

Throughput

Equals offered load × PDR within 0.002 kbps

OLSR overhead

Matches N × 30 HELLOs (every 2 s) plus TC messages

Source drops

0 for AODV and DSDV, which buffer packets

Conclusion and future work

Findings

- AODV delivered the most packets: 77.9% vs 66.2% (OLSR) and 65.0% (DSDV)
- Its advantage is largest under mobility: 75% vs about 50% at 15 m/s
- The cost is about twice the routing overhead and a higher delay
- Single runs ranged from 0% to 100%: repeated, paired runs are essential

Future work

- Tune protocol timers and add a warm-up period
- More concurrent flows and TCP traffic
- More runs per setting and a fading radio model
- Energy use and black hole attack resilience

Thank you — questions?